

VERSION: 1-1

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UNIVERSITY OF TORONTO

FACULTY OF APPLIED SCIENCE AND ENGINEERING

Multiple Choice Assessment

November 2, 2021

First Year

APS110 - ENGINEERING CHEMISTRY AND MATERIALS SCIENCE

Exam Type: B (Duration 1.5 hours. Closed book. Aid sheet provided.)

Examiner(s): F Gu, CQ Jia, SD Ramsay

1. Which of the following is a reasonable justification for the use of the 0.2% offset yield strength?
 - (a) This offset represents exactly the point where mild steel begins to undergo uniform plastic deformation.
 - (b) It is difficult to determine from experimental data where uniform plastic deformation begins.
 - (c) It is difficult to determine from experimental data where non-uniform plastic deformation begins.
 - (d) None of these are reasonable.

Correct answers: (b)

2. Two rectangular cross-section beams are prepared from the same ceramic material. The first beam has length 78 mm, height 9.1 mm, and width 18 mm, while the second beam has length 51 mm, height 9.4 mm, and width 12 mm. The first beam is found to fracture when the applied load, in three-point bending, is 2520 N. What load will the second beam fracture at?
 - (a) 1735.4 N
 - (b) 1195.1 N
 - (c) 1156.9 N
 - (d) 2741.6 N

Correct answers: (d)

3. A hypothetical metal alloy is loaded in tension to just before fracture, at which point the strain is observed to be 0.03. The sample is then unloaded completely and the strain is found to be 0.02. If the Young's modulus of this alloy is 143 GPa, what is the fracture stress of this alloy?
 - (a) 1430.0 MPa
 - (b) 2.86 MPa
 - (c) 4.29 MPa
 - (d) 4290.0 MPa

Correct answers: (a)

4. A hypothetical FCC metal has a density of $6.0 \frac{g}{cm^3}$, a molar mass of $85.53 \frac{g}{mol}$ and an energy for vacancy formation of $0.86 \frac{eV}{atom}$. The Boltzmann constant is $8.617 \times 10^{-5} \frac{eV}{K}$. Which of the following is the correct number of vacancies in this metal at equilibrium at $928^\circ C$?

- (a) $1.04e+19 \frac{vacancies}{cm^3}$
- (b) $9.05e+17 \frac{vacancies}{cm^3}$
- (c) $1.71e+26 \frac{vacancies}{cm^3}$
- (d) $8.92e+20 \frac{vacancies}{cm^3}$

Correct answers: (a)

5. Which of the following strengthening mechanisms would you expect to change the Young's modulus of a metal?

- (a) None of these.
- (b) Minor alloying additions (solid solution).
- (c) Grain size reduction.
- (d) Cold working.

Correct answers: (a)

6. An hypothetical metal has a close packed crystal structure, a density of $8.0 \frac{g}{cm^3}$ and a molar mass of $63.1 \frac{g}{mol}$. Which of the following is the correct number of atoms per unit volume for this metal?

- (a) $9.54e+21 \frac{atoms}{cm^3}$
- (b) $7.63e+22 \frac{atoms}{cm^3}$
- (c) $8.38e-22 \frac{atoms}{cm^3}$
- (d) $1.27e-01 \frac{atoms}{cm^3}$

Correct answers: (b)

7. An hypothetical ceramic has the sodium chloride crystal structure. The anion and cation radii are 0.199 nm and 0.101 nm, respectively. The anion and cation molar masses are $25.29 \frac{g}{mol}$ and $45.45 \frac{g}{mol}$, respectively. Which of the following is the theoretical density of this ceramic?

- (a) $2.18 \frac{g}{cm^3}$.
- (b) $1.09 \frac{g}{cm^3}$.
- (c) $0.54 \frac{g}{cm^3}$.
- (d) $4.35 \frac{g}{cm^3}$.

Correct answers: (a)

8. Which of the following is the correct coordination number for atoms within the BCC crystal structure?

- (a) 4.
- (b) 8.
- (c) 12.
- (d) 6.

Correct answers: (b)

9. Which of the following is necessarily true on the stress strain curve for a typical ductile metal?

- (a) The yield strength must be higher than the proportional limit
- (b) The fracture strength must be higher than the yield strength
- (c) The onset of necking must occur at a lower stress than the fracture strength
- (d) The beginning of uniform plastic deformation must be lower than the fracture strength

Correct answers: (a)

10. Which of the following statements is false?

- (a) An alloying element addition that forms a compound with the solvent atoms is an example of a three dimensional imperfection.
- (b) A minor alloying element addition forming a solid solution is an example of a point imperfection.
- (c) A grain boundary can be characterized as an interfacial imperfection.
- (d) A vacancy is an example of a dislocation.

Correct answers: (d)

11. A hypothetical metal alloy is loaded in tension to a stress of 740 MPa, at which point the strain is observed to be 0.02. The sample is then unloaded completely and the strain is found to be 0.01. What is the Young's modulus of this alloy?

- (a) 74.0 GPa
- (b) 12210.0 GPa
- (c) 37.0 GPa
- (d) 74.0 GPa

Correct answers: (a)

12. A cylindrical cross-section tensile specimen is loaded such that the stress in the reduced section is 461 MPa. If the diameter of the reduced section and the grip region are 28.7 mm and 47.5 mm, respectively, what is the stress in the grip region?

- (a) 1262.8 MPa
- (b) 7994.1 MPa
- (c) 278.5 MPa
- (d) 168.3 MPa

Correct answers: (d)

13. Assuming there exist more than two possible energy states for a hypothetical population of atoms, which of the following statements is true of the distribution of atoms over possible energy states for this population of atoms at infinitely high temperature?
- (a) Only the highest and the lowest energy states will be populated.
 - (b) Only the lowest energy state will be populated.
 - (c) Only the highest energy state will be populated.
 - (d) All of the possible energy states will be equally populated.

Correct answers: (d)

14. A hypothetical metal tensile specimen with a Young's modulus of 110 GPa has an initial length of 37900 mm and is loaded in tension beyond the yield strength. The sample is then unloaded completely and is found to have an unloaded length of 39300 mm. The sample is then loaded again until its loaded length is 39500 mm. What is the stress that the sample is loaded to on this second cycle?
- (a) 6.04 MPa
 - (b) 754.62 MPa
 - (c) 5821.88 MPa
 - (d) 6036.94 MPa

Correct answers: (b)

15. Two cylindrical cross-section tensile specimens are produced from the same material. Sample A has a diameter of 16.9 mm while sample B has a diameter of 40.6 mm. If both samples are loaded by a force of 1600 N and sample A elongates by 535 mm, what will be the elongation of sample B?
- (a) 92.9 mm
 - (b) 222.9 mm
 - (c) 1284.0 mm
 - (d) 3081.6 mm

Correct answers: (a)

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